

NETWORKS AND COMPLEXITY

Solution 15-7

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 15.7: aSIS steady states [Lvl. 3]

Compute the steady states of the aSIS model, studied already in Ex. 15.6. This will require good ‘navigational skills’. If you get lost check the solution.

Solution

Let’s also find the nontrivial steady states. We need to solve

$$0 = p[SI] - r[I] \quad (1)$$

$$0 = r[SI] - 2p[SS][SI]/[S] + w[SI] \quad (2)$$

$$0 = p[SI] + p[SI]^2/[S] - 2r[II]. \quad (3)$$

Before we even start solving it, let’s see if there is a way to use our knowledge about the trivial state to make the search for non-trivial states easier. Have you spotted it already? In the second equation, all terms are linear in $[SI]$, so unless $[SI] = 0$, we can divide the whole equation by $[SI]$. At this point we are pretty sure that the trivial solution that we guessed above is the only steady state where $[SI] = 0$, so we can divide the equation without losing any solutions and replace the second equation with

$$0 = r + w - 2p \frac{[SS]}{[S]}. \quad (4)$$

The key to dealing with equation systems of intermediate complexity is staying in control and not letting them grow into a mess. Because we don’t like fractions, we multiply this equation by $[S]$. The system now reads

$$0 = p[SI] - r[I] \quad (5)$$

$$0 = r[S] - 2p[SS] + w[S] \quad (6)$$

$$0 = p[SI] + p[SI]^2/[S] - 2r[II], \quad (7)$$

where I have also collected the $[II]$ terms in the third equation. We can now multiply the third equation by $[S]$ to clear the remaining fraction, which yields

$$0 = p[SI] - r[I] \quad (8)$$

$$0 = r[S] - 2p[SS] + w[S] \quad (9)$$

$$0 = p[SI][S] + p[SI]^2 - 2r[II][S], \quad (10)$$

Since we only have a single $[II]$, let’s eliminate it using the link conservation law in the form

$$[II] = \frac{z}{2} - [SS] - [SI] \quad (11)$$

after we substitute this, the third equation reads

$$0 = p[SI][S] + p[SI]^2 - 2r(z/2 - [SS] - [SI])[S] \quad (12)$$

$$= (p + 2r)[SI][S] + p[SI]^2 - rz[S] + 2r[SS][S] \quad (13)$$

It seems everything will be nicer if we multiply this equation by another p . After that, our system reads

$$0 = p[SI] - r[I] \quad (14)$$

$$0 = r[S] - 2p[SS] + w[S] \quad (15)$$

$$0 = (p + 2r)p[SI][S] + p^2[SI]^2 - rzp[S] + 2p[SS][S]r \quad (16)$$

Now we can write the second equation as

$$2p[SS] = (w + r)[S] \quad (17)$$

and substitute. The result is

$$0 = p[SI] - r[I] \quad (18)$$

$$0 = (p + 2r)p[SI][S] + p^2[SI]^2 - prz[S] + r(w + r)[S]^2 \quad (19)$$

Using the first equation to replace $p[SI]$ with $r[I]$ gives

$$0 = (p + 2r)r[I][S] + r^2[I]^2 - prz[S] + r(w + r)[S]^2 \quad (20)$$

Since all link variables have now disappeared from the equation we can simplify the notation by dropping the brackets

$$0 = (p + 2r)rSI + r^2I^2 - przS + r(w + r)S^2 \quad (21)$$

Of course we have to take care that we remember that these S and I are still in normalized units, so we should not confuse them with the variables that we defined in the beginning of the chapter, but this risk is rather small compared to the risk of making an algebraic mistake when manipulating long equations. In any case, we are almost done. Let's use the node conservation law $S = 1 - I$ to eliminate S from the equation.

$$0 = (p + 2r)r(1 - I)I + r^2I^2 - prz(1 - I) + r(w + r)(1 - I)^2 \quad (22)$$

$$= (p + 2r)rI - (p + 2r)rI^2 + r^2I^2 - prz + przI + r(w + r) - 2r(w + r)I + r(w + r)I^2 \quad (23)$$

$$= (-pr - 2r^2 + r^2 + r^2 + rw)I^2 + (pr + 2r^2 - 2r^2 - 2rw + prz)I + r(w + r - pz) \quad (24)$$

$$= r(w - p)I^2 + r(p - 2w + prz)I + r(w + r - pz) \quad (25)$$

$$= (w - p)I^2 + (p - 2w + pz)I + (w + r - pz) \quad (26)$$

$$(27)$$

Now it's a good idea to give the prefactors names

$$a = w - p \quad (28)$$

$$b = p(1 + z) - 2w \quad (29)$$

$$c = w + r - pz \quad (30)$$

which allows us to write the polynomial in the standard form

$$0 = aI^2 + bI + c. \quad (31)$$

The solution is the ‘quadratic formula’

$$I = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{32}$$

which is the answer we are looking for.